Gift of the Givers · Relief Management System

DELIVERABLE 03

Application Flow

Site map, user journeys, state machines and route inventory, traced from the screen to the database.

DOCUMENT	VERSION	DATE	STATUS
D-03	2.0	2026-11-02	Current — reflects the system after audit remediation

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1. Site map

Public (anonymous)

Home	/ → intro once per session
About · live impact figures	/Home/About
Relief projects	/Home/ReliefProjects
Donate → PayFast	/Home/Donate
Receipt (by reference)	/Donate/Confirmation
Volunteer application	/Home/Volunteer
Contact	/Home/Contact
Privacy notice (POPIA)	/Home/Privacy
Register (always Donor) · Login	/Identity/Account/*
PayFast ITN webhook (server)	POST /Payments/Notify
Health probes	/health · /health/live

Donor role

Dashboard (confirmed totals)	/Donor/Dashboard
My donations + status	/Donor/MyDonations
Donation details	/Donor/DonationDetails/{id}
Certificate*	/Donor/TaxCertificate/{id}
Certificate PDF*	/Donor/DownloadTaxCertificate
Account, personal data	/Identity/.../Manage

* also available to Employees for any verified donation
Legacy /Donate/Tax-Certificate/{id} redirects here

Employee role (granted by staff)

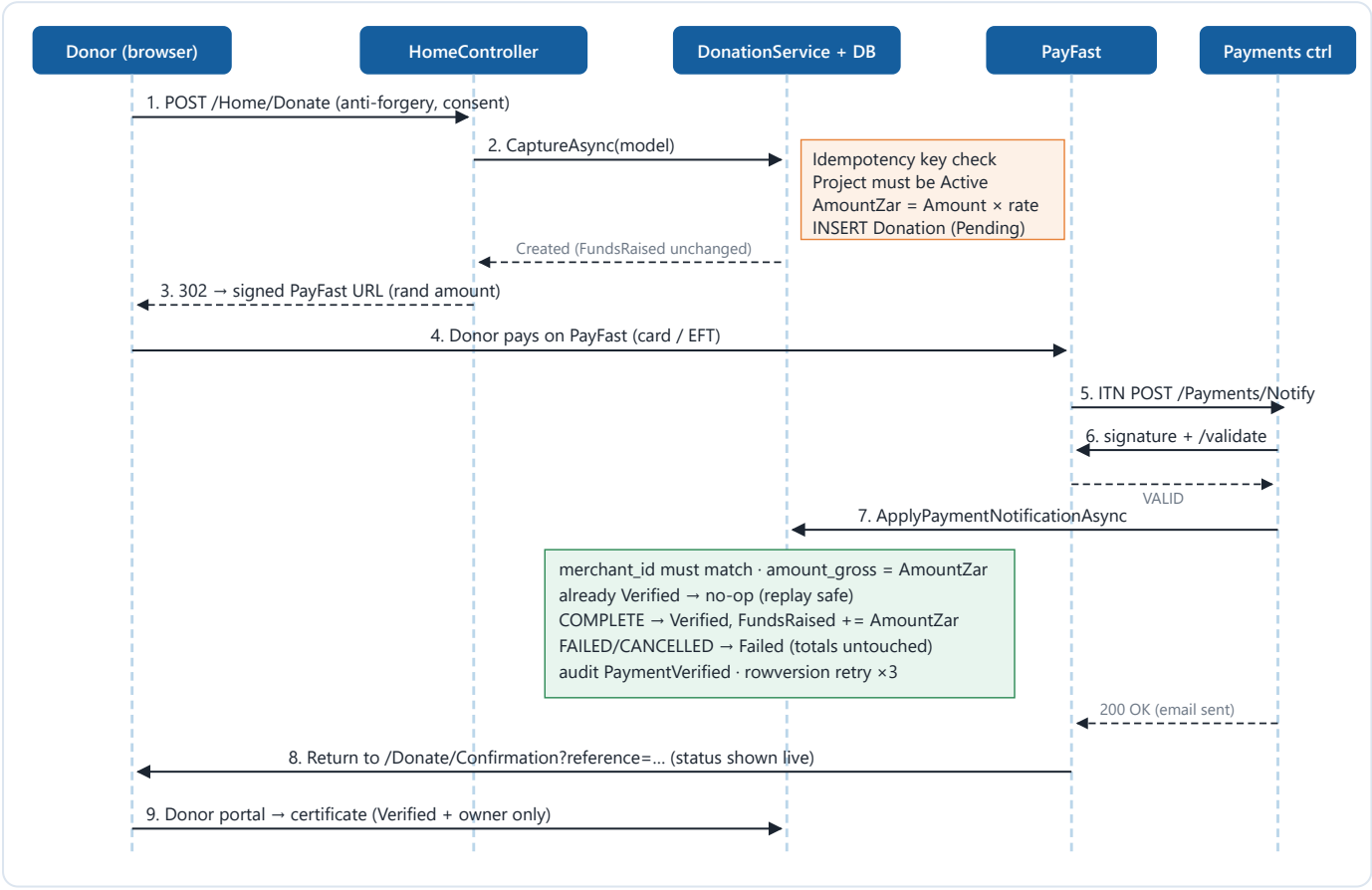
Dashboard + attention cards	/Employee/Dashboard
Projects · create/edit · updates	/Employee/ReliefProjects
Volunteers (filter, paged)	/Employee/Volunteers
Approve / reject / assign	/Employee/VolunteerDetails
Donations (status filter) · CSV	/Employee/Donations
Funding reconciliation	/Employee/FundingReconciliation
Inquiries	/Employee/Inquiries
Staff access (grant/revoke)	/Employee/StaffAccess
Audit log	/Employee/AuditLog

Site map by role — every page and the role that can reach it

Navigation rules:

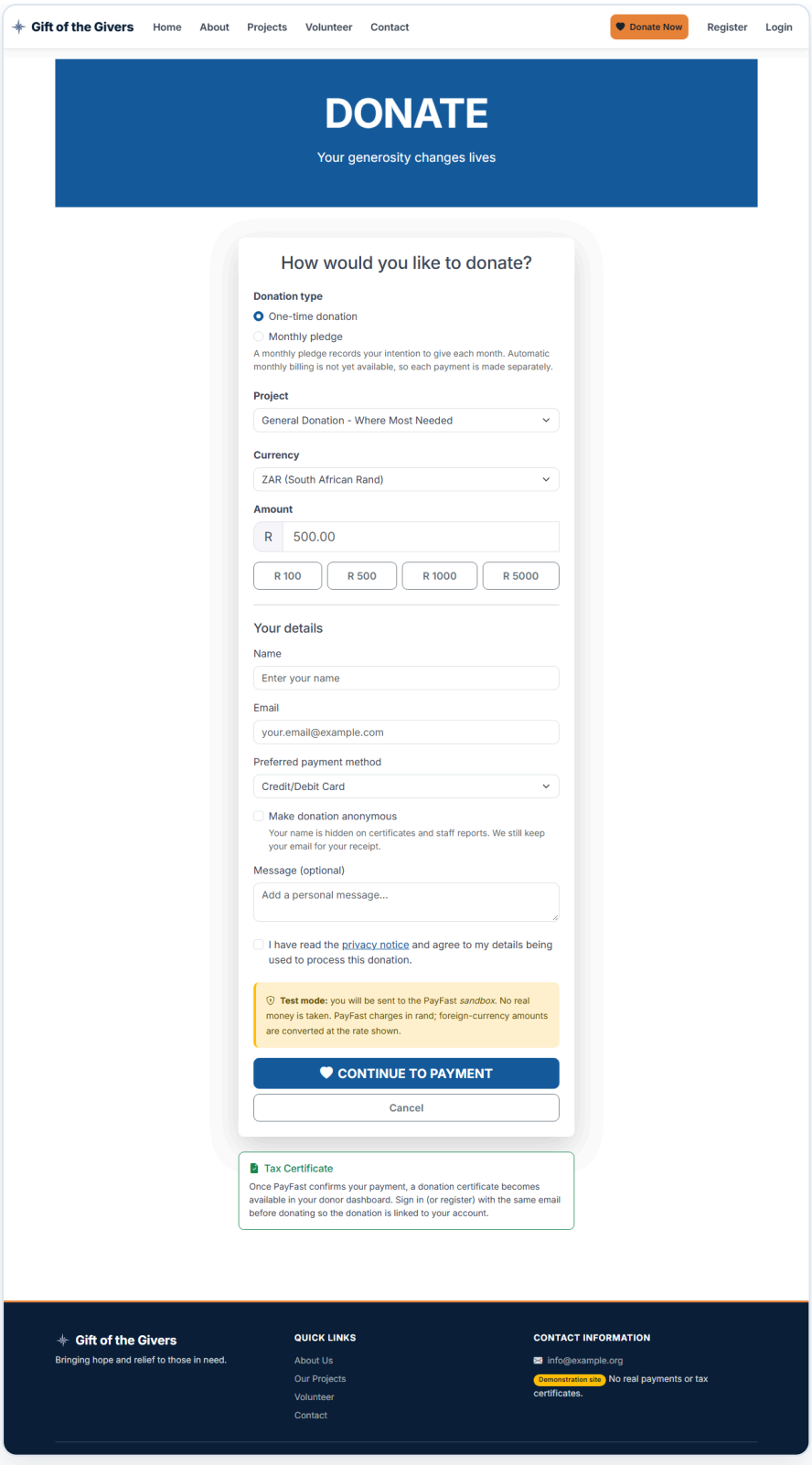
- The top navigation always shows Home, About, Projects, Volunteer, Contact and the **Donate Now** button.
- Signed-in donors also see *Donor Portal*. Employees see *Employee Portal*: Dashboard, Projects, Volunteers, Donations, Funding Reconciliation, Inquiries, Staff Access, Audit Log.
- On the first visit to `/` in each browser session, the visitor is sent to a short intro animation (skippable, and reduced-motion aware), which then returns them to the home page.

2. Journey J1 — Donate (core journey)

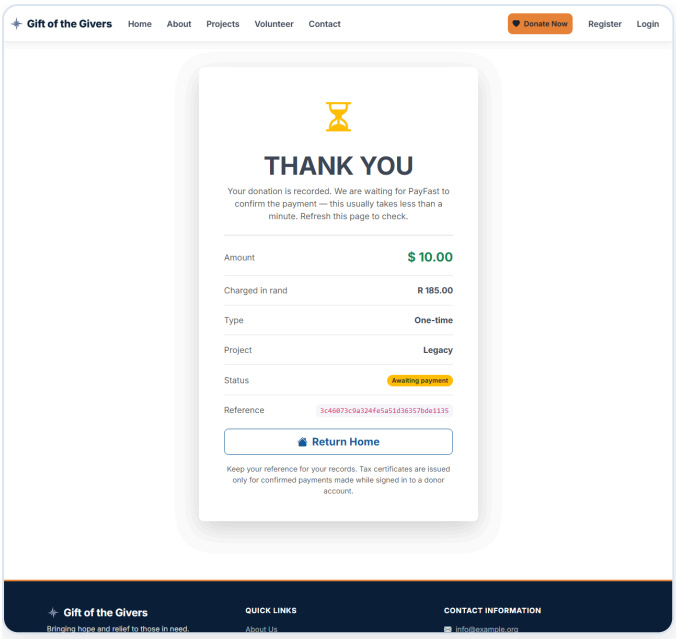


Donation and payment verification sequence

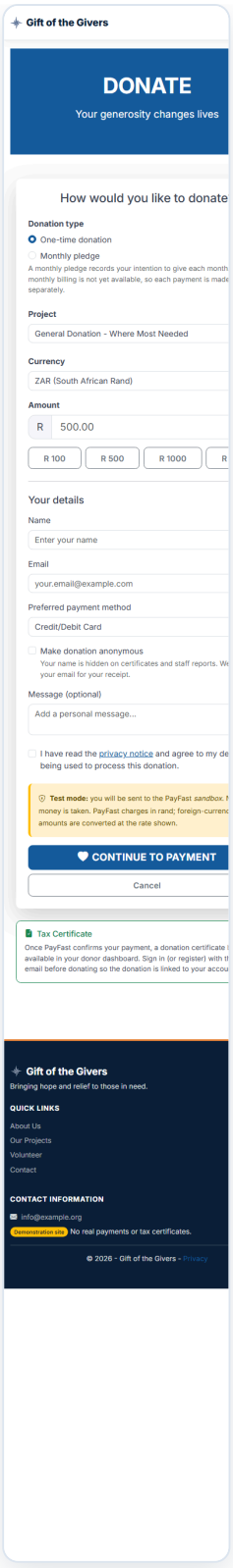
Step	Screen / route	System behaviour	Data
1	`GET /Home/Donate?projectId=`	Form shown. Signed-in donors get their name and email pre-filled. Indicative rand equivalent shown for USD/EUR	—
2	`POST /Home/Donate`	Anti-forgery, validation (type, currency, amount 1–1,000,000, payment method, consent). Errors re-display the form with a summary	—
3	`DonationService.CaptureAsync`	Duplicate token → same donation. Project must be Active. Rand conversion. Insert as **Pending**	`Donations` +1 row; **project total unchanged**
4	302 to PayFast	Signed URL for the rand amount; return, cancel and notify URLs carry the reference	—
5	PayFast	Donor pays by card or EFT, or cancels	—
6	`POST /Payments/Notify` (server to server)	Signature → validate → merchant → amount → status	`Donations.PaymentStatus`, `ReliefProjects.FundsRaised`, `AuditLogs`
7	`GET /Donate/Confirmation?reference=`	Receipt with live status (Awaiting payment / Paid / Failed)	—
8	Donor portal	Certificate available once Paid (owner only)	—



The donation form (desktop)



Receipt while the payment is pending (USD 10 charged as R 185.00)



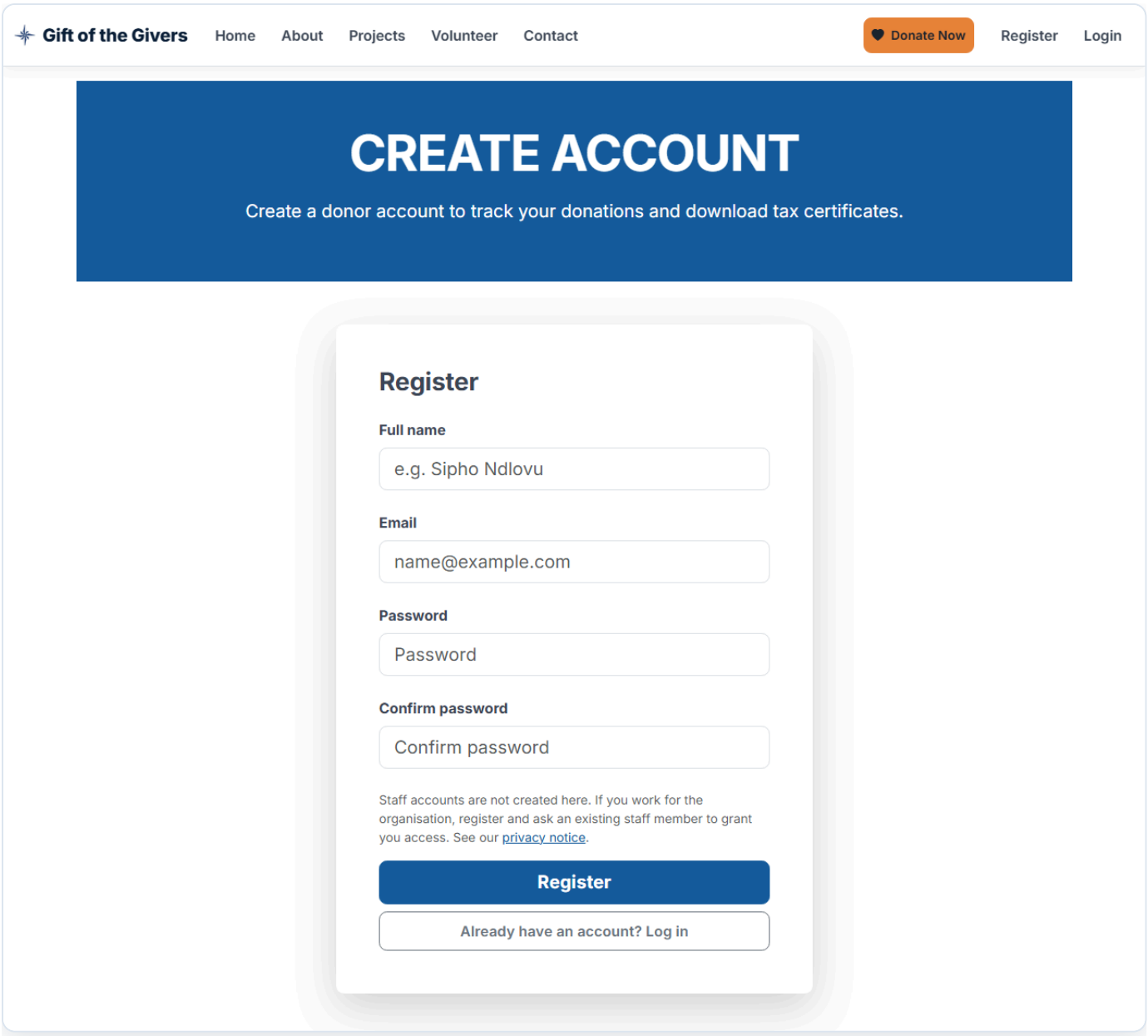
Donate on a 390 px wide phone

Failure and edge paths

Situation	Behaviour
Invalid input (e.g. unknown currency, consent not ticked)	Form re-displayed with errors (HTTP 200); nothing stored
Project suspended or completed	"Not accepting donations" message; nothing stored
Double click / resubmit	Same donation returned (idempotency key); the second redirect goes to the same checkout
Donor cancels on PayFast	Back to Donate with a "payment cancelled, not charged" banner; donation stays Pending
PayFast reports FAILED	Donation marked Failed; totals untouched; receipt offers *Try again*
ITN replayed	200, no change
ITN amount or merchant mismatch	400, donation stays Pending, error logged
Payments disabled (`PayFast:Enabled=false`)	Donation stored as Pending; receipt explains that nothing was charged
More than 20 POSTs per minute from one IP	429 Too Many Requests

3. Journey J2 — Register and staff provisioning

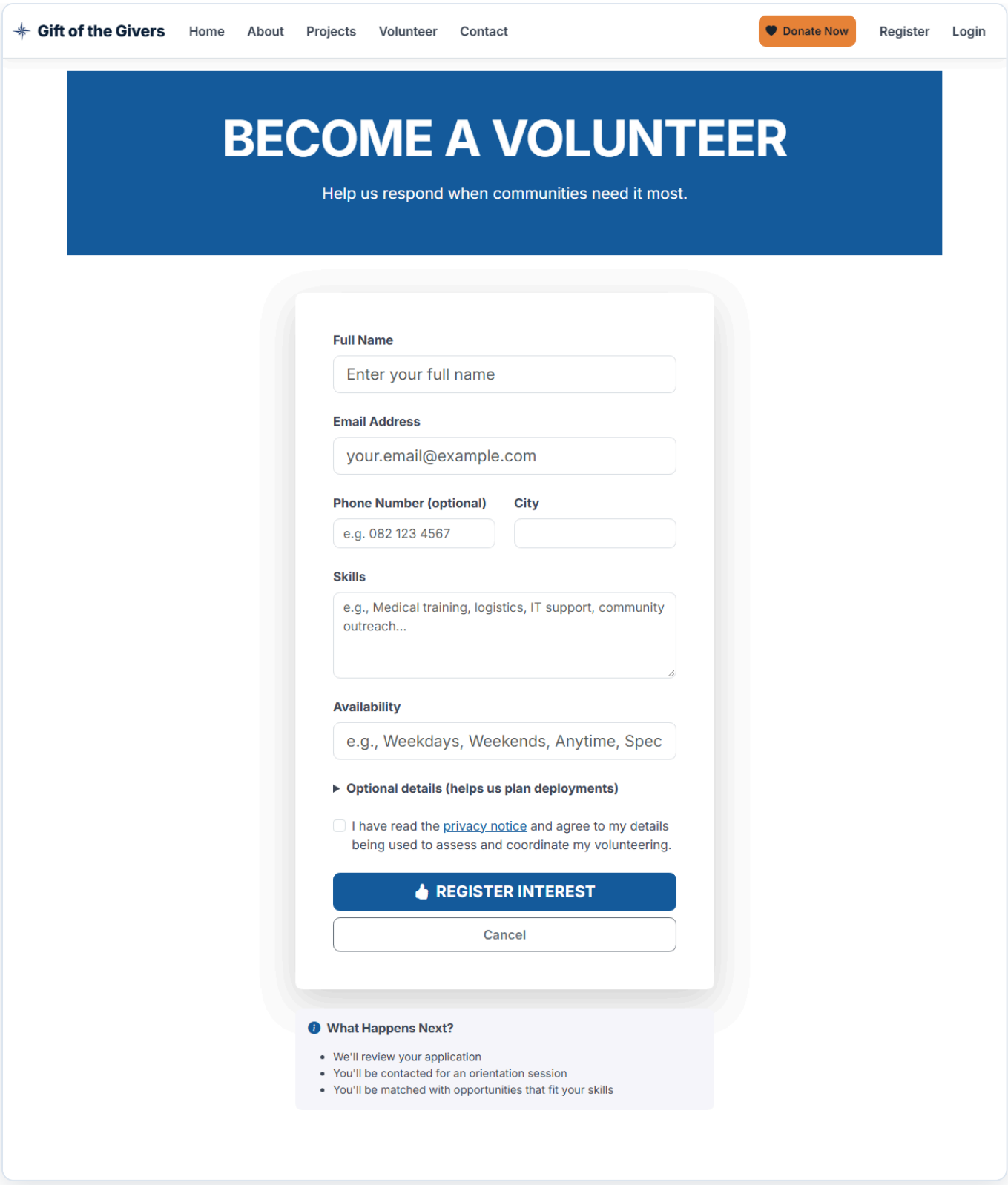
Step	Actor	Route	Result
1	Visitor	`GET/POST /Identity/Account/Register`	Account created with role **Donor** (no role choice). Signed in
2a	First staff member	Deployment sets `StaffAccess:BootstrapEmployeeEmails`	On the next start, the account gets **Employee** (logged)
2b	Existing employee	`GET /Employee/StaffAccess` → `POST GrantEmployee(email)`	Role granted; audit entry `EmployeeGranted`
3	Employee	`POST RevokeEmployee(userId)`	Role removed and security stamp rotated; cannot revoke self; audited



Registration page — no staff option

4. Journey J3 — Volunteer

Step	Actor	Route	Result
1	Visitor	`GET/POST /Home/Volunteer`	Validated (consent required; DOB optional, ≥ 16 years); stored as **Pending** with consent time
2	Employee	`/Employee/Volunteers?status=Pending`	Paged list filtered by status
3	Employee	`/Employee/VolunteerDetails/{id}` → Approve / Reject	Status set, audit entry, email if configured
4	Employee	Assign to an active project / Remove	`VolunteerAssignments` row (unique per pair); audited



Volunteer application form

5. Journey J4 — Donation certificate

Step	Actor	Route	Rule
1	Donor	`/Donor/MyDonations`	Certificate icon only on Paid rows
2	Donor / Employee	`/Donor/TaxCertificate/{id}`	404 unless Verified **and** (owner or Employee); view audited
3	Donor / Employee	`/Donor/DownloadTaxCertificate/{id}`	PDF; download audited; demonstration watermark until PBO configured
—	Legacy links	`/Donate/Tax-Certificate/{id}[/download]`	302 to the routes above
—	Integration	Function `GET /api/tax-certificate/{reference}` + function key	Verified only

6. Journey J5 — Funding reconciliation

1. The employee dashboard shows "*N projects whose funding total needs review*".
2. `/Employee/FundingReconciliation` lists, per project: offline funds, paid online, expected, recorded, difference, and awaiting payment.
3. **Recalculate** sets recorded = offline + paid (audit `FundingRecalculated`).
4. **Accept as offline** keeps the recorded total and moves the unexplained difference into offline funds (audit `FundingAcceptedAsOffline`). Only allowed when the difference is positive.

7. Journey J6 — Contact

`POST /Home/Contact` stores the inquiry with consent → confirmation page → an acknowledgement email if SendGrid is configured → staff see it at `/Employee/Inquiries` (open first) → *Mark handled* (audited).

8. State machines

Donation payment status

From	Event	To	Side effects
—	Form submitted	Pending	None on totals
Pending	ITN COMPLETE, amount and merchant match	Verified	`FundsRaised += AmountZar`; certificate available; email; audit
Pending	ITN with any other status	Failed	Audit only
Verified	Any ITN	Verified	None (idempotent)
Failed	Any ITN	Failed	None

Volunteer status

From	Event	To
—	Application submitted	Pending
Pending / Rejected	Approve	Approved (ApprovalDate set)
Pending / Approved	Reject	Rejected (ApprovalDate cleared)

Project status

`Active` ⇄ `Suspended` → `Completed` (set by staff). Only **Active** projects accept donations and new volunteer assignments.

9. Route inventory

Method	Route	Auth	Purpose
GET	`/`, `/Home/About`, `/Home/ReliefProjects`, `/Home/Privacy`	Public	Information pages
GET/POST	`/Home/Donate`	Public	Donation form / capture → PayFast
GET	`/Donate/Confirmation?reference=`	Public (unguessable reference)	Receipt
GET/POST	`/Home/Volunteer`, `/Home/Contact`	Public	Applications and inquiries
GET	`/Volunteer/Confirmation`, `/Contact/Confirmation`	Public	Confirmations
POST	`/Payments/Notify`	PayFast signature + validation	ITN webhook
GET	`/health`, `/health/live`	Public	Readiness / liveness
*	`/Identity/Account/*`	Varies	Register (Donor only), Login, Manage
GET	`/Donor/Dashboard`, `/Donor/MyDonations`, `/Donor/DonationDetails/{id}`	Donor	Donor portal
GET	`/Donor/TaxCertificate/{id}`, `/Donor/DownloadTaxCertificate/{id}`	Donor (owner) or Employee	Certificates
GET	`/Donate/Tax-Certificate/{id}[/download]`	Donor or Employee	Legacy redirects
GET	`/Employee/Dashboard`, `ReliefProjects`, `Volunteers`, `VolunteerDetails/{id}`, `Donations`, `ExportDonations`, `FundingReconciliation`, `Inquiries`, `AuditLog`, `StaffAccess`	Employee	Staff views
GET/POST	`/Employee/CreateReliefProject`, `EditReliefProject/{id}`, `CreateProjectUpdate`, `EditProjectUpdate/{id}`	Employee	Project management

Method	Route	Auth	Purpose
POST	`/Employee/ApproveVolunteer`, `RejectVolunteer`, `AssignVolunteer`, `UnassignVolunteer`, `MarkInquiryHandled`, `RecalculateFunding`, `AcceptFundingAsOffline`, `GrantEmployee`, `RevokeEmployee`	Employee	Staff actions (audited)
GET	Function `/api/tax-certificate/{reference}`	Function key	Certificate PDF